1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In breast cancer (TCGA-BRCA), high Tumor-Infiltrating Lymphocyte (TIL) density predicts greater response to anti-PD-1/PD-L1 therapy predominantly in tumors with proficient mismatch repair (MMR) via MSH6 and MLH1, which yields a high burden of clonal neoantigens. Conversely, in tumors with MMR dysfunction (loss-of-function in MSH6 and/or MLH1), despite high TIL density, responses to PD-1/PD-L1 blockade are attenuated due to increased neoantigen heterogeneity and subclonal dominance that reduces effective T cell recognition and antigenic clonality. Therefore, the combination of TIL density and MMR status (MSH6/MLH1) will stratify responders more accurately than TIL density alone. TMB will serve as a supporting biomarker, with clonal versus subclonal neoantigen burden mediating the effect. Secondary modifiers, including HLA-DQB1 genotype (antigen presentation efficiency), TP53 mutation status (tumor immunogenicity and genomic instability), and urine DNA methylation biomarkers (epigenetic state), will modulate the strength of this interaction by altering antigen presentation and immune recognition. This yields testable predictions: (1) among TIL-high tumors, MMR-proficient cases show higher therapy_response rates than MMR-deficient cases; (2) among MMR-proficient tumors, higher clonal neoantigen load correlates with better response; (3) secondary modifiers shift the magnitude of the TIL/MMR interaction. The endpoints will be objective therapy_response metrics (e.g., RECIST-based ORR/longitudinal progression-free survival) in cohorts treated with anti-PD-1/PD-L1 agents.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, TMB
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 and TMB because they connect a specific gene-level MMR lesion to the quantity and potentially the clonality/quality of neoantigens that determine T cell recognition.

TMB: TMB/MSI), some bladder tumors with dense tumor-infiltrating lymphocytes (TILs) fail to respond to PD-1 blockade — why does immune infiltration not translate into effective immunotherapy response in these tumors? 2) Tier 1 paths used: I built on the mechanistic trio linking MMR/TMB (MSH6 → altered mismatch repair → elevated TMB/MSI) and tissue immune context (PD-1/TIL axis).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	HLA-DQB1	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	urine-DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	GARS1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: as MMR status determinants, (c) a mutation- and clonality-based neoantigen framework (TMB and clonality), and (d) endpoint therapy_response to anti-PD-1/PD-L1 therapy. Secondary modifiers (HLA-DQB1, TP53, urine DNA methylation biomarkers) are biologically plausible regulators of antigen presentation and tumor immunogenicity and may explain residual variability in response.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- HLA-DQB1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.